SafeHer

Complete Developer Setup Guide

Fresh-machine handover: clone GitHub, configure credentials,
run FastAPI in VS Code, run Android in Android Studio,
and connect the app to the backend.

Repository: https://github.com/Ghige/SAFEHER
Document date: 2026-10-05

NO SECRETS IN THIS DOCUMENT

Source of truth: docs/SAFEHER_DEVELOPER_SETUP.md
Do not use example LAN IPs as permanent values.
Do not commit .env, local.properties, google-services.json, or keystores.

Complete fresh-machine handover for cloning GitHub, configuring credentials, running the FastAPI backend in VS Code, running the Android app in Android Studio, and connecting the two.

- Repository (verified remote): https://github.com/Ghige/SAFEHER.git
- Guide date: 2026-10-05
- Scope: documentation and configuration templates only. This guide does not redesign the app, AI, SOS, auth, APIs, models, or thresholds.
- Companion PDF: docs/SAFEHER_DEVELOPER_SETUP.pdf (regenerate with python scripts/generate_developer_setup_pdf.py)

This Markdown file is the complete source of truth. The PDF is the printable copy of the same guide.

Status labels used in this guide

Table: Label / Meaning

VERIFIED | Confirmed from files in this repository
NOT VERIFIED | Could not be confirmed from the repository; needs a developer/environment check
REQUIRES USER CREDENTIAL | A teammate must provide a private file or secret
REQUIRES EXTERNAL ACCOUNT | Needs Firebase / Google / Meta / Twilio / Postgres access
REQUIRES MANUAL SETUP | Must be done on the machine; Git clone is not enough

1. Project overview

SafeHer is a personal-safety platform:

- Android app (safeher_android/): Kotlin + Jetpack Compose, package com.safeher.app, versionName 2.0.0, versionCode 1 (VERIFIED in app/build.gradle.kts).
- Backend (safeher_backend/): FastAPI app app.main:app, JWT auth, PostgreSQL, optional Redis, Firebase Admin, Meta WhatsApp Cloud API.
- On-device AI: SafeHer V2 audio pipeline bundled in the APK (VERIFIED on main after commit 0b975b6): yamnet_embedding.tflite then safeher_classifier_v2.tflite. AiModelVersion.AUDIO_DISTRESS = "v2.0". Live path is confirm-first in local uncommitted work; whether that Stage 3 code is on GitHub is a known blocker (see section 4). AI_AUTO_SOS_ENABLED=false on Android and backend when those flags exist. Manual SOS is separate.
- Web track page (safeher_web/track/): static emergency-contact map page. Backend can serve it at /track/ when the monorepo layout is intact.
- Training pipeline (safeher_ai/): V1 YAMNet fine-tune scripts plus V2 embedding/classifier scripts under safeher_ai/training/*_v2.py. Not required to run the app if the V2 model files are present (they are tracked in Git).

A developer cloning this repository should not retrain the model just to run the application.

2. Architecture overview (as implemented)

```
Physical phone or emulator
        |
        |  HTTP(S) to FastAPI
        v
safeher_android  ---- Retrofit ---->  safeher_backend (uvicorn :8000)
        |                                   |
        | TFLite in assets                  | PostgreSQL (required)
        | AudioMonitorService (mic FGS)     | Redis (optional in development)
        | LocationForegroundService         | Firebase Admin JSON (FCM)
        | Firebase (google-services.json)   | Meta WhatsApp (optional for local login tests)
        v
Confirm-first AI popup. Auto SOS is OFF.
```

Debug API URL selection (VERIFIED in DebugApiBaseUrl.kt + app/build.gradle.kts):

1. Release builds always use https://api.safeher.app/. DEBUG_API_BASE_URL is compiled as empty. Release forbids cleartext (app/src/main/res/xml/network_security_config.xml).

2. Debug builds with DEBUG_API_BASE_URL set in gitignored local.properties: that LAN URL is used for all debug devices, including emulators.

3. Debug builds with no override:

- Emulator: http://10.0.2.2:8000/ (special alias to the host machine).

- Physical device: Gradle debug BASE_URL also falls back to http://10.0.2.2:8000/ when the override is empty, which does not work on a real phone. For a physical phone you must set DEBUG_API_BASE_URL to your laptop LAN IP.

Debug network security allows cleartext HTTP (app/src/debug/res/xml/network_security_config.xml).

CURRENT MACHINE EXAMPLE ONLY -- DO NOT USE AS A PERMANENT VALUE:

```
http://192.168.1.17:8000/
```

Every developer must look up their own IPv4 on the Wi-Fi that the phone uses. DHCP addresses change.

3. Project structure (actual repository)

```
SAFEHER/
??? README.md                          # Points here and to the PDF
??? PRODUCTION_SETUP.md                # Production DNS / Firebase / WhatsApp checklist
??? TEAM_HANDOVER.pdf                  # Short older onboarding PDF
??? docs/
?   ??? SAFEHER_DEVELOPER_SETUP.md     # This file
?   ??? SAFEHER_DEVELOPER_SETUP.pdf    # Printable copy
??? scripts/
?   ??? generate_developer_setup_pdf.py
?   ??? generate_handover_pdf.py
??? .github/workflows/ci.yml           # Backend pytest on Python 3.11 + Postgres 16
??? safeher_android/                   # Android Studio project root
?   ??? app/                           # Application module
?   ??? gradle/wrapper/                # Gradle 9.3.0 wrapper
?   ??? local.properties.example
?   ??? keystore.properties.example
??? safeher_backend/                   # FastAPI project (open this folder in VS Code)
?   ??? app/                           # Application code (main.py, api, models, services)
?   ??? alembic/versions/              # Migrations 001-008
?   ??? tests/
?   ??? requirements.txt
?   ??? .env.example
?   ??? .env.production.example
?   ??? docker-compose.p0.yml          # Optional Redis + optional Postgres
?   ??? firebase-credentials.json.example
??? safeher_ai/                        # Training / export; models/final holds TFLite
??? safeher_web/                       # Static /track page
??? PLAY_STORE_*.md                    # Store checklists (not required to run locally)
```

Table: Folder / Role

safeher_android/ | Open this directory in Android Studio.
safeher_backend/ | Open this directory in VS Code. Run uvicorn from here.
safeher_ai/ | Train/export YAMNet. Not needed for day-to-day app run.
safeher_web/ | Emergency contact track HTML.
docs/ | This handover.
.github/workflows/ | CI: backend tests with OTP_PROVIDER=mock.

NOT VERIFIED: exact list of every untracked local-only file on a given developer machine.

4. KNOWN SETUP BLOCKERS

Do not treat a Git clone as a one-click install. The following must be completed or acknowledged.

1. REQUIRES USER CREDENTIAL / REQUIRES EXTERNAL ACCOUNT: Android Gradle Google Services plugin is applied (libs.plugins.google.services). A real safeher_android/app/google-services.json is required to sync/build. It is gitignored. Placeholders in google-services.json.example will not talk to Firebase. Obtain the file from a teammate or Firebase Console.

2. REQUIRES USER CREDENTIAL: Backend .env is gitignored. Copy .env.example. PostgreSQL must exist and match DATABASE_URL.

3. REQUIRES MANUAL SETUP: Android local.properties is gitignored. Android Studio usually writes sdk.dir. Physical-device testing also needs DEBUG_API_BASE_URL.

4. REQUIRES EXTERNAL ACCOUNT (optional for login-only local tests): Firebase Admin JSON at FIREBASE_CREDENTIALS_PATH is required for backend FCM. Login/register against Postgres can work without it; push notifications will not. NOT VERIFIED whether every backend boot path fails hard if the JSON is missing.

5. REQUIRES MANUAL SETUP: Physical phone and laptop must share Wi-Fi. Windows Firewall must allow inbound TCP 8000. Uvicorn must bind 0.0.0.0.

6. WhatsApp / Twilio / Maps: Not required to register/login with mock OTP. Required for real SMS OTP, maps tiles, and emergency WhatsApp. REQUIRES EXTERNAL ACCOUNT.

7. GitHub vs this working tree: A clone of origin/main after this documentation push includes SafeHer audio V2 assets (yamnet_embedding.tflite + safeher_classifier_v2.tflite) from commit 0b975b6. This machine still has a large uncommitted Android/backend working tree (Stage 3 confirm-first, DEBUG_API_BASE_URL, Gradle syncTfliteAsset for the older yamnet_distress_quant.tflite, etc.). A colleague who only clones GitHub will not automatically get that uncommitted work. DEBUG_API_BASE_URL is documented from the local tree; if DebugApiBaseUrl.kt is missing on the clone, physical-device LAN override will not work until that code is committed.

8. Android asset / Gradle mismatch (local dirty tree): Merged main deleted safeher_android/app/src/main/assets/yamnet_distress_quant.tflite and requires V2 files instead. Uncommitted local app/build.gradle.kts may still try to copy yamnet_distress_quant.tflite. Do not assume Gradle on GitHub matches the dirty local Gradle file.

9. Clean-machine install: This guide was not executed on a blank PC as an end-to-end proof. Commands were checked against repository files, not against a fresh OS image.

10. Release signing: keystore.properties and *.jks are gitignored. Debug builds do not need them. Play Store / release APK requires them (REQUIRES USER CREDENTIAL).

11. Training datasets: safeher_ai/data/ is gitignored and not required to run inference.

5. Prerequisites (install first)

REQUIRED

Table: Tool / Version / notes / Status

Git | Any current Git for Windows | VERIFIED needed to clone
Python | 3.11.x (comment in safeher_backend/requirements.txt; CI uses 3.11) | VERIFIED
PostgreSQL | 16 used in CI and docker-compose.p0.yml. Other 14+ versions NOT VERIFIED | VERIFIED for 16 in CI/compose
Android Studio | README previously said Ladybug+. Exact installer version NOT VERIFIED | NOT VERIFIED
JDK | Android compileOptions / kotlinOptions target JVM 17. Android Studio bundled JBR 21 can compile that target. | VERIFIED target 17
Android SDK | Platform 35 (compileSdk = 35, targetSdk = 35), minSdk = 26 | VERIFIED
Gradle | Wrapper 9.3.0 (gradle-wrapper.properties) | VERIFIED
Android Gradle Plugin | 8.7.3 (gradle/libs.versions.toml) | VERIFIED

Kotlin | 2.0.21 | VERIFIED
GitHub access | Clone https://github.com/Ghige/SAFEHER.git (private access may be required) | REQUIRES EXTERNAL ACCOUNT

OPTIONAL

Table: Tool / Why
Redis 7 | Dead-man switch + rate limits. Dev falls back to in-memory if Redis is down (redis_service.py). Production/staging refuse that fallback when REQUIRE_REDIS_IN_PRODUCTION=true.
Docker Desktop | safeher_backend/docker-compose.p0.yml can start Redis and (profile with-db) Postgres.
Meta WhatsApp Cloud API | Emergency contact messages.
Twilio Verify | Real SMS OTP (OTP_PROVIDER=twilio).
Google Maps API key | Map tiles. Empty MAPS_API_KEY still builds.
VS Code Python extension | Convenience only.

DEVELOPMENT ONLY

- OTP_PROVIDER=mock and OTP_ALLOW_DEV_SHORTCUT=true (defaults in .env.example).
- Debug cleartext HTTP to LAN / 10.0.2.2.
- ENVIRONMENT=development, DEBUG=true.

PRODUCTION ONLY (do not use for this local guide)

- ENVIRONMENT=production, OTP_PROVIDER=twilio, OTP_ALLOW_DEV_SHORTCUT=false.
- Release base URL https://api.safeher.app/.
- Redis required. See PRODUCTION_SETUP.md and .env.production.example.

SDK components (VERIFIED as required by Gradle; exact SDK Manager checkbox names NOT VERIFIED):

- Android SDK Platform 35
- Android SDK Build-Tools (version not pinned in Gradle; Android Studio installs a default)
- Android SDK Platform-Tools (adb)
- Google Play services / Google APIs as pulled by Gradle dependencies
- Emulator image: x86_64 or ARM system image with Google APIs, API 26+ (API 35 recommended). Exact AVD name NOT VERIFIED.

6. Developer credential checklist

REQUIRED before a useful local run

- GitHub access to this repository
- Python 3.11
- PostgreSQL running with a database matching DATABASE_URL
- Backend .env copied from .env.example (placeholders replaced)
- SECRET_KEY changed from the example string for anything beyond throwaway local use
- Android Studio + JDK 17-capable toolchain
- Android SDK Platform 35
- safeher_android/local.properties with valid sdk.dir (Studio usually creates this)
- Real safeher_android/app/google-services.json (not the .example file)
- Emulator or physical Android 8.0+ (API 26+) device
- Runtime grants: microphone, notifications, location (when using SOS/maps/AI)

REQUIRED for physical-phone + backend

- Laptop and phone on the same Wi-Fi
- Laptop LAN IPv4 (not 10.0.2.2, not 127.0.0.1)
- DEBUG_API_BASE_URL=http://<YOUR_LAN_IP>:8000/ in local.properties

- Uvicorn --host 0.0.0.0 --port 8000
- Windows Firewall inbound TCP 8000 allowed

REQUIRED for FCM / backend push

- Firebase project access
- Backend firebase-credentials.json (Admin SDK) and FIREBASE_CREDENTIALS_PATH

OPTIONAL

- Redis
- MAPS_API_KEY
- Meta WhatsApp tokens
- Twilio Verify credentials
- Release keystore (keystore.properties + JKS)

PRODUCTION ONLY (not this guide)

- Production .env from a secret store
- Approved WhatsApp templates
- DNS for api.safeher.app / safeher.app

7. SAFE SECURITY RULES

NEVER COMMIT

- Passwords and database URLs with real passwords
- API keys (OpenAI, Google, Maps, Firebase client keys, Meta, Twilio)
- Private keys and PEM files
- Firebase Admin / service-account JSON (firebase-credentials.json, *adminsdk*.json, *service-account*.json)
- Production database credentials
- JWT SECRET_KEY values
- Filled .env files
- local.properties (machine SDK path + optional LAN URL + Maps key)
- google-services.json (contains a Google API key; gitignored)
- Signing keystores (*.jks, *.keystore) and keystore.properties
- User recordings / safeher_ai/data/ datasets

Root .gitignore already ignores .env, local.properties, google-services.json, firebase-credentials.json, venvs, build/, and safeher_ai/data/.

DO COMMIT (templates only)

- safeher_backend/.env.example
- safeher_backend/.env.production.example
- safeher_backend/firebase-credentials.json.example
- safeher_android/local.properties.example
- safeher_android/keystore.properties.example
- safeher_android/app/google-services.json.example

Never paste real secret values into this Markdown, the PDF, README, or chat logs.

8. Configuration files table

Table: File / Required? / Used by / Where to put it / How to obtain it / Commit to Git?

File: safeher_backend/.env -- Required?: Yes (runtime) -- Used by: Backend -- Where to put it: safeher_backend/.env -- How to obtain it: Copy .env.example; fill locally -- Commit to Git?: NO

File: safeher_backend/.env.example -- Required?: Template -- Used by: Humans / this guide -- Where to put it: Already in repo -- How to obtain it: In Git -- Commit to Git?: YES
File: safeher_backend/.env.production.example -- Required?: Template -- Used by: Production ops -- Where to put it: Already in repo -- How to obtain it: In Git -- Commit to Git?: YES
File: safeher_backend/firebase-credentials.json -- Required?: Required for FCM; optional for login-only -- Used by: Firebase Admin -- Where to put it: Path in FIREBASE_CREDENTIALS_PATH (default ./firebase-credentials.json) -- How to obtain it: Firebase Console: Project settings > Service accounts > Generate new private key -- Commit to Git?: NO
File: safeher_backend/firebase-credentials.json.example -- Required?: Template -- Used by: Humans -- Where to put it: Already in repo -- How to obtain it: In Git -- Commit to Git?: YES
File: safeher_android/local.properties -- Required?: Yes -- Used by: Gradle -- Where to put it: safeher_android/local.properties -- How to obtain it: Copy example; Studio writes sdk.dir -- Commit to Git?: NO
File: safeher_android/local.properties.example -- Required?: Template -- Used by: Humans -- Where to put it: Already in repo -- How to obtain it: In Git -- Commit to Git?: YES
File: safeher_android/app/google-services.json -- Required?: Yes (Gradle Google Services + FCM) -- Used by: Android -- Where to put it: safeher_android/app/google-services.json -- How to obtain it: Firebase Console download, or teammate via a private channel -- Commit to Git?: NO
File: safeher_android/app/google-services.json.example -- Required?: Template -- Used by: Humans -- Where to put it: Already in repo -- How to obtain it: In Git -- Commit to Git?: YES
File: safeher_android/keystore.properties -- Required?: Release only -- Used by: Gradle signing -- Where to put it: safeher_android/keystore.properties -- How to obtain it: Copy example; private keystore -- Commit to Git?: NO
File: yamnet_embedding.tflite -- Required?: Yes (on-device V2) -- Used by: Android assets -- Where to put it: safeher_android/app/src/main/assets/ and safeher_ai/models/v2/ -- How to obtain it: Already in Git -- Commit to Git?: YES
File: safeher_classifier_v2.tflite -- Required?: Yes (on-device V2) -- Used by: Android assets -- Where to put it: same -- How to obtain it: Already in Git -- Commit to Git?: YES
File: yamnet_distress_quant.tflite -- Required?: Legacy V1 artifact -- Used by: Training + backend copy -- Where to put it: safeher_ai/models/final/, safeher_backend/app/ai/audio_model/ -- How to obtain it: Still tracked; removed from Android assets on main (V2 commit) -- Commit to Git?: YES (legacy)
File: yamnet_distress.tflite -- Required?: Optional legacy -- Used by: Training artifacts -- Where to put it: safeher_ai/models/final/ -- How to obtain it: In Git -- Commit to Git?: YES
File: Alembic versions 001-006 -- Required?: Yes on GitHub -- Used by: DB schema -- Where to put it: safeher_backend/alembic/versions/ -- How to obtain it: In Git -- Commit to Git?: YES
File: Alembic 007, 008 -- Required?: Local only today -- Used by: Uniqueness / AI feedback -- Where to put it: same folder -- How to obtain it: Uncommitted on this machine -- Commit to Git?: NO (not on main yet)
File: safeher_backend/docker-compose.p0.yml -- Required?: Optional -- Used by: Redis / Postgres -- Where to put it: safeher_backend/ -- How to obtain it: In Git -- Commit to Git?: YES
File: CI secrets -- Required?: CI only -- Used by: GitHub Actions -- Where to put it: Workflow env in .github/workflows/ci.yml uses dummy SECRET_KEY and test DB -- How to obtain it: In Git (dummy values) -- Commit to Git?: YES (dummy only)

google-services.json is not safe to publish on GitHub in this project: it is gitignored and contains a restricted Google API key. Share it privately. The .example file is placeholders only.

Observed local Firebase project id (from a private local file, not committed): safeher-4521a. Teammates need Console access to that project or their own Firebase project with matching Android application ids. REQUIRES EXTERNAL ACCOUNT.

oauth_client arrays in the observed local google-services.json were empty. SHA-1 / SHA-256 for Google Sign-In are NOT VERIFIED as required for the current phone+password/OTP login path.

9. Backend setup (VS Code)

9.1 Open the project

1. Install VS Code.
2. File > Open Folder > select safeher_backend (not the monorepo root, unless you prefer a multi-root workspace).

3. Select Python 3.11 as the interpreter after the venv exists.

9.2 Virtual environment (Windows PowerShell)

Commands below match README / CI style. README uses .venv inside safeher_backend. Either .venv or venv311 is fine; this guide uses .venv as the documented default.

```
cd safeher_backend
py -3.11 -m venv .venv
.\.venv\Scripts\activate
python -m pip install --upgrade pip
pip install -r requirements.txt
```

If py -3.11 is missing, install Python 3.11 from python.org and retry. Wrong major versions are a common ModuleNotFoundError / wheel source.

requirements.txt pins FastAPI 0.115.6, uvicorn 0.34.0, SQLAlchemy 2.0.36, Alembic 1.14.0, pytest 8.3.4, firebase-admin, redis, psycopg2-binary, numpy 1.26.4.

9.3 Configure .env

```
copy .env.example .env
```

Then edit .env. Do not invent production passwords. Placeholders from .env.example:

```
DATABASE_URL=postgresql://postgres:password@localhost:5432/safeher_db
SECRET_KEY=change-this-to-a-random-secret-key
ALGORITHM=HS256
ACCESS_TOKEN_EXPIRE_MINUTES=1440
REDIS_URL=redis://localhost:6379
ENVIRONMENT=development
DEBUG=true
OTP_PROVIDER=mock
OTP_ALLOW_DEV_SHORTCUT=true
AI_AUTO_SOS_ENABLED=false
FIREBASE_CREDENTIALS_PATH=./firebase-credentials.json
CORS_ALLOWED_ORIGINS=
```

- DATABASE_URL user/password/db must match the Postgres role you actually created. The example uses user postgres, password password, db safeher_db, host localhost, port 5432. Those are template values, not a guarantee your machine uses them.
- Optional Docker compose Postgres uses user safeher, password safeher_dev_only_change_me, db safeher_db (see section 10).
- Development CORS is * when ENVIRONMENT=development (VERIFIED in app/main.py).
- TrustedHostMiddleware is off unless production-like (_PROD).

9.4 Firebase Admin (backend)

1. Firebase Console > Project settings > Service accounts > Generate new private key.
2. Save as safeher_backend/firebase-credentials.json (gitignored).
3. Leave FIREBASE_CREDENTIALS_PATH=./firebase-credentials.json or set an absolute path.
4. Do not commit this file. The .example file is a shape-only placeholder.

9.5 Start the API

Local-only testing (browser/Swagger on the same PC; Android emulator can still use 10.0.2.2 which maps to the host). Default uvicorn bind without --host is 127.0.0.1:

```
python -m uvicorn app.main:app --reload --port 8000
```

Physical Android device testing (phone must reach the laptop). Bind all interfaces:

```
python -m uvicorn app.main:app --reload --host 0.0.0.0 --port 8000
```

README also documents:

```
uvicorn app.main:app --reload --host 0.0.0.0 --port 8000
```

(equivalent if the venv Scripts directory is on PATH).

Verify Swagger

- Open http://127.0.0.1:8000/docs
- Liveness: GET http://127.0.0.1:8000/api/v1/ping/ should return "status": "ok" and "ai_auto_sos_enabled": false
- Readiness: GET http://127.0.0.1:8000/api/v1/ping/ready requires PostgreSQL (database=ok or HTTP 503)

Do not send a real emergency alert while verifying setup.

10. Database setup (PostgreSQL)

Alembic reads DATABASE_URL from settings (alembic/env.py). alembic.ini has an empty sqlalchemy.url; do not put passwords there.

There is no automatic user seed. Register a user through the app or POST /api/v1/auth/register. Model-version seed exists only for AI dataset admin endpoints (ensure_seed_model_versions) and is not required for login.

Option A -- local PostgreSQL (manual)

Install PostgreSQL 16 (version 16 VERIFIED in CI). In psql or pgAdmin, create a database. Example using the template credentials from .env.example (replace if your install uses a different postgres password):

```
CREATE DATABASE safeher_db;
```

If you create a dedicated user (optional; do not invent production passwords):

```
CREATE USER safeher WITH PASSWORD '<YOUR_PASSWORD>';
GRANT ALL PRIVILEGES ON DATABASE safeher_db TO safeher;
```

Then set:

```
DATABASE_URL=postgresql://<YOUR_USER>:<YOUR_PASSWORD>@localhost:5432/safeher_db
```

Windows default superuser is often postgres. The password is whatever you chose at install time (REQUIRES MANUAL SETUP).

Option B -- Docker (optional)

From safeher_backend:

```
docker compose -f docker-compose.p0.yml --profile with-db up -d
```

Compose values (VERIFIED in docker-compose.p0.yml):

- user: safeher
- password: safeher_dev_only_change_me (dev-only sample in compose; change if you expose the port)
- database: safeher_db
- port: 5432

Matching URL:

```
DATABASE_URL=postgresql://safeher:safeher_dev_only_change_me@localhost:5432/safeher_db
```

Redis-only:

```
docker compose -f docker-compose.p0.yml up -d
```

Migrations (required)

From safeher_backend with venv active and .env loaded:

```
alembic upgrade head
```

Tracked revisions on GitHub main (VERIFIED via git ls-files):

Table: Revision file / Purpose

001_create_users.py | users
002_create_alerts.py | alerts / contacts related schema
003_create_helpers.py | helpers
004_add_user_is_verified.py | is_verified
005_add_user_fcm_token.py | FCM token column
006_ai_dataset_and_model_versions.py | AI dataset tables

Local uncommitted files 007_uniqueness_alerts_contacts.py and 008_ai_feedback_events.py exist in this working tree but are not on GitHub main. A clone will only have 001--006 until those are committed. KNOWN BLOCKER if the unpublished backend code expects 007/008 tables.

CI uses database safeher_test with user/password safeher -- that is for GitHub Actions only.

11. Firebase / Google configuration

Android application ids (VERIFIED)

Table: Variant / Application id

Release | com.safeher.app
Debug | com.safeher.app.debug (applicationIdSuffix = ".debug")

Register both Android apps in Firebase if you run debug installs.

google-services.json

- Required at safeher_android/app/google-services.json
- Used by the Google Services Gradle plugin and Firebase (FCM, Crashlytics plugin is applied)
- Do not commit
- Download from Firebase Console for this project, or receive it through a private team channel
- Copying google-services.json.example to google-services.json is not sufficient for a working Firebase connection

Backend Admin SDK

- File: firebase-credentials.json (private key -- never commit)
- Env: FIREBASE_CREDENTIALS_PATH
- Needed for server-side FCM

Auth in this codebase

App login is phone + password via FastAPI (/api/v1/auth/register, /login, /send-otp, /verify-otp). It is not Firebase Auth email/password. Firebase on Android is used for messaging (and Crashlytics), not as the primary user table.

SHA-1 / SHA-256

NOT VERIFIED as required for current OTP/password login. Add debug/release SHA fingerprints in Firebase if you later enable Google Sign-In or restricted API keys that demand them.

Google Maps

MAPS_API_KEY in local.properties is injected as BuildConfig.MAPS_API_KEY and a manifest placeholder. Empty key: app still builds; maps tiles fail.

12. Android setup (Android Studio)

12.1 Open the project

Open the folder safeher_android, not the monorepo root (unless you add it as a module later). Wait for Gradle sync.

12.2 local.properties

```
cd safeher_android
copy local.properties.example local.properties
```

Set:

```
sdk.dir=C:\\Users\\<YourUser>\\AppData\\Local\\Android\\Sdk
MAPS_API_KEY=<YOUR_KEY_OR_LEAVE_EMPTY>
```

Android Studio usually overwrites/creates sdk.dir automatically.

Physical device extra line:

```
DEBUG_API_BASE_URL=http://<YOUR_LAPTOP_LAN_IPV4>:8000/
```

CURRENT MACHINE EXAMPLE ONLY -- DO NOT USE AS A PERMANENT VALUE: http://192.168.1.17:8000/

Find your IPv4 (Windows PowerShell):

```
Get-NetIPAddress -AddressFamily IPv4 | Where-Object { $_.PrefixOrigin -ne 'WellKnown' } | Fo
rmat-Table IPAddress, InterfaceAlias
```

Use the Wi-Fi adapter address (often 192.168.x.x or 10.x.x.x). Do not use 127.0.0.1 (that is the phone itself). Do not use 10.0.2.2 on a physical phone (that alias exists only inside the Android emulator).

After changing DEBUG_API_BASE_URL, Rebuild the app so BuildConfig regenerates.

12.3 google-services.json

Place the real file at safeher_android/app/google-services.json. Gradle sync fails or Firebase init fails without it.

12.4 Sync / build

In Android Studio: File > Sync Project with Gradle Files, then Run.

CLI (from safeher_android, JDK on PATH):

```
.\gradlew.bat assembleDebug
```

Unit tests:

```
.\gradlew.bat testDebugUnitTest
```

JAVA_HOME should point at JDK 17+ (Android Studio JBR is acceptable). NOT VERIFIED that every machine already has JAVA_HOME set.

12.5 Permissions (VERIFIED in AndroidManifest.xml)

Install-time / declared:

- INTERNET, ACCESS_NETWORK_STATE
- ACCESS_FINE_LOCATION, ACCESS_COARSE_LOCATION, ACCESS_BACKGROUND_LOCATION
- RECORD_AUDIO
- POST_NOTIFICATIONS
- FOREGROUND_SERVICE, FOREGROUND_SERVICE_LOCATION, FOREGROUND_SERVICE_MICROPHONE
- REQUEST_IGNORE_BATTERY_OPTIMIZATIONS

Runtime (grant on device): microphone, notifications (API 33+), location (and background location if the OS prompts for SOS tracking).

Foreground services:

- LocationForegroundService -- foregroundServiceType=location
- AudioMonitorService -- foregroundServiceType=microphone

12.6 Emulator vs physical device

See sections 16--18 of the run procedure.

USB debugging on a physical phone: Settings > Developer options > USB debugging. Authorize the PC RSA prompt. adb devices should show device, not unauthorized or offline.

13. AI / SafeHer model setup (as implemented on GitHub main)

Table: Item / Value / Status

Current Android audio pipeline | YAMNet embedding TFLite + SafeHer classifier V2 | VERIFIED in AiModelAssets.kt on merged main
Embedding model | yamnet_embedding.tflite | VERIFIED
Classifier model | safeher_classifier_v2.tflite | VERIFIED
Model version string | AiModelVersion.AUDIO_DISTRESS = "v2.0" | VERIFIED
Bundled in APK? | Yes, under safeher_android/app/src/main/assets/ | VERIFIED
Downloaded at runtime? | No | VERIFIED
Retrain required to run app? | No | VERIFIED
Training dataset required for inference? | No | VERIFIED
Movement TFLite | Not a required RC1 asset | VERIFIED
Auto SOS | AI_AUTO_SOS_ENABLED=false when those flags are present in the checkout | VERIFIED in local/backend templates; confirm Android BuildConfig on the clone
Legacy V1 quant model | yamnet_distress_quant.tflite still in safeher_ai/models/final/ and safeher_backend/app/ai/audio_model/; deleted from Android assets by V2 | VERIFIED
Keras checkpoints | safeher_ai/models/v2/*.keras tracked | VERIFIED

Python training extras (safeher_ai/requirements_training.txt and V2 scripts 01_prepare_dataset_v2.py ... 06_verify_classifier_v2_tflite.py) are optional. On-device inference uses Android TensorFlow Lite 2.16.1 (libs.versions.toml) if that catalog is on the clone.

Do not replace the V2 TFLite files as part of setup.

Local uncommitted note: this workspace may still contain Stage 3 files that load AiModelAssets.AUDIO_MODEL / yamnet_distress_quant.tflite. That is not what a clean GitHub clone of merged main uses. Treat it as unpublished work.

Logcat: filter the app package com.safeher.app.debug. V2 inference logs live in TfliteAudioInferenceEngine. Tag SAFEHER_AI exists only if Stage 3A observability was committed.

14. Complete run procedure (fresh machine)

STEP 1 -- Install prerequisites

Install Git, Python 3.11, PostgreSQL 16, Android Studio (SDK 35). Optional: Redis, Docker Desktop.

STEP 2 -- Clone repository

```
git clone https://github.com/Ghige/SAFEHER.git
cd SAFEHER
```

If the repository is private, authenticate with GitHub (browser, PAT, or SSH). NOT VERIFIED which auth method your org requires.

STEP 3 -- Open backend in VS Code

Open folder SAFEHER\safeher_backend.

STEP 4 -- Create Python virtual environment

```
py -3.11 -m venv .venv
.\.venv\Scripts\activate
```

STEP 5 -- Install dependencies

```
python -m pip install --upgrade pip
pip install -r requirements.txt
```

STEP 6 -- Configure .env

```
copy .env.example .env
```

Set DATABASE_URL and SECRET_KEY. Keep AI_AUTO_SOS_ENABLED=false, OTP_PROVIDER=mock, OTP_ALLOW_DEV_SHORTCUT=true for local mock OTP.

STEP 7 -- Configure PostgreSQL

Create database safeher_db (or match your URL). Then:

```
alembic upgrade head
```

STEP 8 -- Configure Firebase / backend credentials

Place Admin SDK JSON if you need FCM. Skip only if you accept no push from backend.

STEP 9 -- Start backend

Emulator / same-PC:

```
python -m uvicorn app.main:app --reload --port 8000
```

Physical phone:

```
python -m uvicorn app.main:app --reload --host 0.0.0.0 --port 8000
```

STEP 10 -- Verify Swagger

Browse http://127.0.0.1:8000/docs and http://127.0.0.1:8000/api/v1/ping/.

STEP 11 -- Open Android project in Android Studio

Open SAFEHER\safeher_android.

STEP 12 -- Configure local.properties

Copy the example; confirm sdk.dir. For a physical phone, set DEBUG_API_BASE_URL to your LAN URL.

STEP 13 -- Configure google-services.json

Place the real file in safeher_android/app/.

STEP 14 -- Sync Gradle

Android Studio: Sync Project with Gradle Files.

STEP 15 -- Build Android project

Run app (debug) or .\gradlew.bat assembleDebug.

STEP 16 -- Run emulator

Create an AVD (API 26+, Google APIs). Cold boot. Run the app configuration. Default debug URL http://10.0.2.2:8000/ reaches uvicorn on the PC if DEBUG_API_BASE_URL is unset. If you set a LAN override, the emulator will also use that LAN IP; the PC must listen on 0.0.0.0.

STEP 17 -- Run physical Android device

Enable USB debugging, accept RSA, adb devices = device. Install the debug variant (com.safeher.app.debug).

STEP 18 -- Configure physical-device backend URL

1. Discover LAN IPv4 on the laptop.
2. Set DEBUG_API_BASE_URL=http://<LAN>:8000/ (trailing slash recommended; Gradle normalizes).
3. Rebuild/reinstall.
4. Start uvicorn with --host 0.0.0.0 --port 8000.
5. Confirm phone Wi-Fi (not mobile data only, unless you have a verified routing path -- NOT VERIFIED for cellular).

STEP 19 -- Test registration / login / profile

1. Register with a phone number + password (API POST /api/v1/auth/register).
2. Send OTP, then verify. With mock + shortcut, OTP 123456 is accepted (otp_service.py, OTP_ALLOW_DEV_SHORTCUT=true).
3. Login (POST /api/v1/auth/login).
4. Open Profile in the app.

Password rules: follow in-app validation (unit tests mention length checks). Exact UX copy: NOT VERIFIED beyond code.

STEP 20 -- Test SafeHer AI

1. Grant microphone + notifications.
2. Start AI / audio monitoring in the app (AudioMonitorService).
3. Confirm the foreground notification appears.
4. You should not get an automatic SOS (AI_AUTO_SOS_ENABLED=false). Confirm-first popup may appear on sustained distress; dismiss/confirm as designed.
5. Do not play loud distress audio in public without warning.

STEP 21 -- Test SOS flow

1. Grant location.
2. Add at least one emergency contact if the UI requires it for dispatch.
3. Use manual SOS if you need to exercise /api/v1/alerts/trigger.
4. Real WhatsApp delivery needs Meta credentials -- otherwise expect backend logs without a live message. Do not send a real emergency to real contacts.

STEP 22 -- Troubleshooting

See section 15.

15. Troubleshooting

Backend / Python

Table: Symptom / What to check

ModuleNotFoundError | Venv not activated; pip install -r requirements.txt not run; opened wrong folder
Wrong Python | python --version must be 3.11.x
pip / wheel failure | Use 3.11, upgrade pip, avoid 3.13 unless NOT VERIFIED
Postgres connection failure | Service running; DATABASE_URL user/password/db/port; alembic upgrade head; ping/ready
Firebase Admin errors | Missing JSON or bad FIREBASE_CREDENTIALS_PATH

Port 8000 in use | Another uvicorn; change port and URLs together

Android / Gradle

Table: Symptom / What to check
Gradle sync failure | Network for Maven; JDK; wrapper 9.3.0 download
Missing google-services.json | File must be exactly app/google-services.json
Android SDK missing | SDK Manager: Platform 35
sdk.dir missing | local.properties
TFLite missing | Tracked model should copy via syncTfliteAsset; check assets

Device / network

Table: Symptom / What to check
Device not detected | USB cable, drivers, adb kill-server then adb devices
unauthorized | Unlock phone, accept debugging RSA
offline | Replug, revoke USB debugging authorizations
Network error in app | Backend down; wrong base URL; cleartext blocked (should not happen on debug)
Emulator cannot reach API | Use 10.0.2.2 or LAN override + 0.0.0.0; not 127.0.0.1 inside emulator unless that is intended
Phone cannot reach API | 10.0.2.2 does not work on a physical phone. Use laptop LAN IP. Phone and PC on same Wi-Fi. Uvicorn 0.0.0.0. Windows Firewall inbound 8000. Disable guest-Wi-Fi AP isolation if the AP blocks client-to-client.
Windows Firewall | Allow Python/uvicorn TCP 8000 inbound on Private networks

10.0.2.2 = emulator alias for the host loopback.

Physical phone = developer laptop LAN IPv4.

Permissions / AI

Table: Symptom / What to check
No mic capture | RECORD_AUDIO denied; OEM permission drawer
No notification | POST_NOTIFICATIONS on API 33+
FGS killed | Battery optimization; FOREGROUND_SERVICE_MICROPHONE
TFLite load failure | Asset missing; Logcat TfliteAudioInferenceEngine
AI never auto-alerts | Expected: auto SOS is off

Logcat filter examples:

```
package:com.safeher.app.debug
tag:SAFEHER_AI
tag:okhttp.OkHttpClient
```

16. Auth, OTP, and CORS (local)

- Register/login: FastAPI JWT (SECRET_KEY, ALGORITHM=HS256).
- Mock OTP: stored code or 123456 when OTP_ALLOW_DEV_SHORTCUT=true.
- Production/staging refuse mock OTP (production_guards.py).
- Dev CORS: allow all origins.
- Do not enable AI_AUTO_SOS_ENABLED for setup.

17. Tests you can run locally

Backend (venv, Postgres not always required for every test -- many tests use the app with env from CI; NOT VERIFIED that every test passes without Postgres on every machine):

```
cd safeher_backend
.\.venv\Scripts\activate
pytest
```

CI equivalent env is in .github/workflows/ci.yml (DATABASE_URL to safeher_test, OTP_PROVIDER=mock, AI_AUTO_SOS_ENABLED=false).

Android:

```
cd safeher_android
.\gradlew.bat testDebugUnitTest
```

This guide does not claim a clean-machine installation was tested unless a later report says so.

18. Related documents

Table: Document / Use

docs/SAFEHER_DEVELOPER_SETUP.pdf | Printable copy of this guide
README.md | Short index
TEAM_HANDOVER.pdf | Older short handover
PRODUCTION_SETUP.md | Production DNS / hosting
safeher_backend/.env.example | Env names
safeher_ai/README.md | Retrain only

19. Verification notes (authoring)

Checked against repository files on 2026-10-05 after merging origin/main (0b975b6 SafeHer V2 audio) with the documentation commit. Gradle versions, SDK levels, package name, .env.example keys, ping routes, OTP shortcut, gitignore, V2 TFLite names, debug/release URL logic in the local tree, compose Postgres sample, CI Python 3.11.

Not executed as a blank-PC install. Android Studio exact version, OEM USB drivers, live Firebase Console IAM, and a clean clone of only GitHub main (without this machine's uncommitted Stage 3 files) were NOT VERIFIED as a full run. Alembic files 007 and 008 were present locally as untracked -- NOT VERIFIED that GitHub main contains them until they are committed.